

EXECUTIVE MEMO

To: Dr. Chris Garcia, Mayo Clinic
From: MITsdm Team 14a, Sanguine
12 May, 2026

Subject: Executive Authorization: Sanguine System Development

Executive Summary

Sanguine is an AI system to improve lab patient care outcomes by augmenting physician-facing test results with relevant research using an agentic AI clinical librarian. Sanguine can exceed competitors' AI features, leverage Mayo Clinic's Platform_Insights data lake and strategic partnership with Google Cloud, and align with the Mayo Clinic ethos by prioritizing clinical accuracy over speed or cost. We propose authorizing \$7.1M to deliver a prototype in Q4 2026, a pilot in Q1 2027, and a US launch with three test types in Q3 2027.

Context

Nowadays, Mayo Clinic processes approximately 26 million laboratory tests annually. These results are delivered solely with statistical reference ranges, without being linked to specific clinical and historical patient data. A comprehensive clinical interpretation relies on the physician's ability to conduct a clinical assessment within the patient's clinical context. Without this clinical context, a gap emerges between the delivery of laboratory results and their clinical interpretation.

Together, patients have increasingly turned to general-purpose artificial intelligence tools to interpret their health information without clinical supervision. At the same time, our diagnostic competitors are introducing AI-assisted support features for result interpretation. At Mayo Clinic, we must ensure that AI-driven interpretation is safe, traceable, and clinically useful, aligned with the Mayo Clinic Model of Care.

This memo introduces Sanguine: a physician-facing agentic AI designed for physicians that enriches laboratory test results with evidence-based, up-to-date, and traceable interpretive support. Sanguine does not replace medical judgment, it does not issue diagnoses or provide interpretations directly to the patient. Sanguine empowers physicians with specialist-grounded research before they act.

Our analysis supports a system model based on specialized medical AI functions that utilizes curated clinical evidence and release controls calibrated according to clinical familiarity and complexity. This approach enables Sanguine to scale clinical review aligned with the Mayo Clinic Model of Care for laboratory results, without human review becoming a bottleneck in the release of results.

We request executive authorization to proceed with Sanguine as an implementation program budgeted at US\$ 7.1 million, spanning 50 weeks and scheduled to start in June 2026. This approval also entails three mandatory institutional commitments, detailed in the Project Plan section: (1) allocation of Mayo Clinic IT team capacity during the Architecture & Definition phase to define and enable controlled access to clinical and laboratory systems, (2) availability of regulatory and institutional capacity also during Architecture & Definition to prepare the pre-submission to the Institutional Review Board (IRB) and (3) reservation of physician reviewer capacity during the Integration and V&V phase to support clinical validation before the pilot deployment.

The problem: scaling the Mayo Clinic Model of Care to laboratory results interpretation

The Mayo Clinic Model of Care is grounded in an integrated, collaborative, and patient-centered experience. Specialists work together across various disciplines, clinical information is interpreted within its specific context, and decisions are based on coordinated judgment rather than isolated data points. Mayo Clinic Laboratories already produces world-class test results, however, the results-presentation layer remains standardized and generic, delivering outcomes as mere numerical values, reference ranges, and basic flags. At the same time, the broader clinical significance still relies on the ordering physician's ability to synthesize the patient's history (including past trends, comorbidities, and medications) alongside specialized knowledge and current scientific evidence.

This creates a systemic scalability problem. Clinical standards demand coordinated, context-specific interpretation, yet, the laboratory workflow operates at such a high volume that Mayo's multidisciplinary model cannot be applied to every individual result. Without additional interpretive support, the Mayo Clinic Model of Care is not applied to laboratory results before physicians make decisions based upon them.

Three premises in the current model that require evaluation:

The issuance of laboratory results is merely a data-delivery function. Laboratory results are not mere data outputs, they are clinical information products intrinsically linked to each patient's condition. If delivered solely as numerical

values, basic alerts, and reference ranges, a significant portion of the clinical value that Mayo can offer remains outside the interpretive workflow, and the cognitive burden of this synthesis is shifted onto the physician.

"The Mayo Clinic Model of Care can be scaled only through human coordination." Mayo's collaborative model succeeds because expertise is coordinated around the patient, leveraging Mayo's resources, including specialists in complementary fields and clinical references derived from the institution's recent academic research. However, applying this same level of contextual synthesis to a high volume of laboratory results demands a new, systemic interpretive layer, rather than merely an intensified manual effort.

"The value of AI lies in answering questions directly." For Sanguine, value is not limited to a response generated by a chatbot. Rather, value resides in a controlled clinical librarian layer, a system that retrieves and compares evidence from various complementary specialties, organizes the patient's context, and assists the physician in interpretation, thereby expanding their clinical capabilities. Therefore, the challenge is not merely to develop an AI capable of providing answers, but also to design an integrated interpretive layer that is clinically valid and operationally scalable. Sanguine addresses this gap by creating a "clinical librarian" layer aimed directly at the physician applied to laboratory results. Its objective is neither to automate diagnoses nor to replace physicians, but rather to make contextual clinical interpretation more scalable by applying the Mayo Clinic Model of Care, bringing relevant patient context, clinical evidence, and specialized synthesis closer to the point of care.

The competitive landscape: competing models for laboratory result interpretation

The competitive landscape is not limited to a comparison of solutions, but rather to a comparison of system models that add clinical value to laboratory results through AI. The approaches taken by Epic, Labcorp, Quest, and Sanguine vary in their support for interpretation, clinical traceability, and integration into medical workflows. Some recent initiatives demonstrate that the market for laboratory result analysis is advancing beyond the mere delivery of results. In February 2026, Labcorp entered into a collaboration with PathAI to conduct digital pathology and AI-assisted diagnostics. Furthermore, its other product, TestFinder, assists physicians in selecting laboratory tests, presented as a menu, directly within their clinical workflows. Sanguine's differentiator is not merely its use of AI, rather, it is the system's architecture: a controlled, multi-agent interpretation layer that applies clinical reasoning guided by the Mayo Clinic Model of Care to laboratory results before the physician takes action.

Epic MyChart	Rule-based result delivery with nascent AI features not yet in general clinical use. No clinical interpretation layer. Physicians bear full cognitive load.
Labcorp and Quest	Single-agent AI with RAG on external sources only and rule-based release control. No patient context integration, no multi-specialist synthesis, no clinical librarian layer.
Sanguine	Multi-specialist agentic synthesis with tiered clinical validation. Mirrors Mayo Model of Care at scale. Focused on serving the role of a clinical librarian to assist physicians

The proposed system: architecture and design rationale

The goal is to improve Mayo Clinic lab patient care outcomes at scale, by augmenting physician-facing test results with relevant research, using an agentic AI clinical librarian. The system boundary is explicitly limited to the interpretive layer between laboratory test results and the physician.

Our proposed system consists of a controlled interpretive layer between laboratory test results and the clinical judgment realized by the physician. **This interpretive layer does not replace the physician**, it neither issues diagnoses nor releases interpretations directly to patients. Instead, it leverages reliable evidence coordinated by autonomous clinical specialists to utilize the patient's context, reconcile different interpretations, and present a traceable summary to support clinical decisions. Clinical validation is performed in shadow mode — outputs are evaluated against specialist physician review before any interpretation reaches a live clinical workflow.

Our design analysis compared different architectures of lower cost and scope against more comprehensive architectures that better preserve clinical reliability. The recommended design features a high reliability architecture that utilizes multiple specialized AI roles, curated evidence sources, controlled boundaries for patient data usage, and a release control mechanism that accounts for clinical recurrence and case complexity. Internally, this design concept is known as "Full," as it prioritizes clinical accuracy, traceability, and readiness for validation.

Architecture Decision	Option	Rationale
Compute Platform	Google Cloud Platform (GCP)	Aligned with Mayo Clinic's multi-year strategic partnership; bundled pricing reduces cost and supports scalable, governed deployment.
Model Origin	Gemini Family	Discounted on GCP; strong multimodal reasoning; no proprietary training overhead; reduces integration complexity by keeping the initial system within one model ecosystem.
Research Paradigm	RAG + Fine-Tuning	Combines external evidence retrieval with domain adaptation; highest clinical accuracy in modeling while grounding outputs in traceable clinical sources.
AI Roles per Result	Large Group (5–10)	Specialist agents for hematology, nephrology, hepatic, etc. mirroring Mayo's collaborative panel.
Data Sources	Multiple Internal + External	Maximizes evidence coverage; cross-source comparison reduces hallucination risk and improves traceability of clinical support.
Release Control	Tiered Release	High-risk, rare or clinically complex results escalate to human review; common results release automatically, balancing accuracy and scale.
Model Specification	Gemini 3 Pro (Multimodal)	Required for lab image integration and complex clinical reasoning across result types.

Design trade-off: balancing cost, scale and clinical assurance

From ten candidate architectures evaluated across six key decisions and stress-tested through Monte Carlo simulation, the “Full” concept emerged as the most robust under uncertainty. The comparison between cost and accuracy highlights the main design trade-off: lower cost architectures, such as the workflow of Epic MyChart or a reference model from Labcorp, require a smaller investment but offer limited accuracy in clinical interpretation. At the other extreme, the Mayo Clinic Model of Care is the "gold standard," as it approaches total clinical assurance by relying on continuous human clinical review, at an annual cost of US\$ 200 million. The proposed concept sits between these two extremes: while not the lowest cost option, it offers accuracy superior to that of its competitors, while avoiding the costs and scalability limitations of the current clinical model.

The uncertainty analysis demonstrated that the highest impact architectural uncertainties are not linked exclusively to the model's capability. Rather, they stem from the manner in which the system grounds its interpretations in clinical evidence, the coordination of specialized AI roles, the way conflicting or incomplete evidence is reconciled, and how release control distinguishes common from those that are rare, ambiguous, or clinically complex. For Sanguine, these mechanisms enable the Mayo Clinic

Architecture	Modeled Accuracy	Annual Cost	Validation
Epic MyChart	25%	\$0 Incremental	Rule-based
Labcorp (competitor)	50%	\$50M	Rule-based
Full (proposed concept)	95%	\$80M	Tiered release
Mayo Clinic Internal (gold standard)	100%	>\$200M	Human in the loop (always)

Model of Care to scale to the interpretation of laboratory results without devolving into a low-reliability system (such as a chatbot) or an unscalable human review process. If those mechanisms are weak, the system could increase patient safety risks, reduce physician trust, and create regulatory exposure. The "Full" concept positions near the Pareto frontier: it is not anchored in the current, fully manual review model, it does not require Mayo to develop and maintain proprietary models from scratch and it avoids single-agent approaches, which offer a lower degree of reliability and more closely resemble solutions from current competitors. Instead, it combines Mayo’s strategic alignment with GCP (Gemini models), a panel of specialized agents with a tiered release control system that forwards only rare and complex cases for human clinical review.

Project plan: implementation path to pilot development

Following the selection of the architecture, we evaluated different implementation approaches, which varied in terms of staffing models, sequencing, risk mitigation activities, costs, and schedule. The recommended plan, already revised to mitigate all identified risks, consists of a 50-week implementation program designed to guide Sanguine from the architecture definition phase through the execution of a private clinical pilot. This plan prioritizes the early definition of data access boundaries, controlled system development, preparation for clinical validation, and integration with physician workflows.

Sanguine will be validated using checkpoints at each phase. It begins with component verifications and advances to integration and system testing, followed by clinical validation in shadow mode before the private pilot. Each phase completes with a review gate to confirm that the package (components and subsystems) is ready to proceed to the subsequent phases.

Phase	Key Delivery and Gate Question
Phase 1 (Jun–Sep 2026): Architecture & Definition	Requirements signed off; data-access scope defined; system boundary, release model, and multi-agent architecture specified. Gate: “Is the system definition complete enough to proceed into component engineering?”
Phase 2 (Oct–Dec 2026): Component Engineering	Data retrieval pipeline; agentic interpretation core (hematology, nephrology, hepatic interpreters); physician output interface. Gate: "Can the system generate a traceable interpretation for representative clinical cases?"
Phase 3 (Jan–Mar 2027): Integration and V&V	Integrated workflow testing; shadow-mode clinical validation; exception testing; physician review feedback. Gate: “Is the system clinically credible, traceable, and safe enough for private pilot use?”
Phase 4 (Apr–Jul 2027): Operational Launch	Private pilot with 1–10 partner providers; US-only limited rollout for 3 selected test types; operational monitoring and regulatory readiness. Gate: “Is the system ready for controlled US launch?”

Risk: prioritizing reliability over cost

Our project risk analysis identified the following 5 risks, which impact Sanguine’s ability to achieve readiness for the clinical pilot. Given that Sanguine operates within a physician-centric clinical workflow, these risks cannot be treated merely as cost or schedule issues. If discovered late in the process, they could block validation, necessitate rework across various parts of the system, delay clinical deployment, or yield results upon which physicians cannot rely. For this reason, the recommended plan mitigates all identified risks rather than prioritizing the lowest cost implementation path. For Sanguine, reliability takes precedence over cost minimization, as failures in clinical accuracy could negatively impact the Mayo Clinic reputation, physician adoption, and, most importantly, patient care. As a result, the set of mitigation actions reduces the project’s overall risk exposure by 58%. This reduction reflects a comprehensive mitigation strategy in which all identified project risks are addressed before they can compromise integration and validation & verification phase.

Risk	Impact	Mitigation	Cost
(EHR) / Lab Integration Failure	High: Blocks all downstream development — no data access means no system to build.	Data integration feasibility spike with Mayo Clinic IT to validate access, interfaces, and data assumptions early.	\$220K
AI Clinical Accuracy Failure	High: Incorrect interpretations create patient safety liability and block regulatory approval for deployment.	Structured clinical validation protocol with a physician expert panel before shadow-mode evaluation.	\$700K
(IRB) / Regulatory Delay	Medium: Approval timelines are unpredictable; a delay of months cascades into the pilot and launch schedule.	IRB pre-submission and regulatory roadmap during the design phase.	\$280K
Patient Data Compliance Failure	Medium: A breach or audit finding could halt the project entirely and	PHI handling review, access controls, audit trail design, and data-minimization review	\$260K

	damage Mayo Clinic brand trust.	before patient data is used.	
Late Interface Rework	High: Components that work independently fail together, forcing expensive rework across all phases simultaneously.	Early integration and interface testing across the data pipeline, retrieval layer, agentic AI core, physician workflow, and validation workflow.	\$420K

Budget, schedule, and scope summary

Item	Value	Note
Total Implementation Budget (with full mitigation)	\$7.10M	Recommended program budget for the 50-week implementation plan
Labor (7 teams, 43 wks)	\$4.69M	Core internal teams plus targeted contractors
Risk Mitigation Work	\$1.88M	Full mitigation set covering all project risks identified in the risk analysis
Implementation Timeline	50 weeks	Architecture definition through Integration & V&V and controlled US operational launch
Prototype Target	Q4 2026	Physician-facing prototype demonstrating traceable interpretation for representative cases
Private Pilot Target	Q1 2027	Pilot with 1–10 partner providers
Cost per interpreted result at 26M tests/yr	<\$10	Target of 26M annual laboratory tests

Call to action

Sanguine’s objective is to extend the Mayo Clinic Model of Care to laboratory results that do not yet receive that level of contextual interpretation at scale. Mayo Clinic Laboratories already has the clinical knowledge, research base, specialist expertise, and institutional trust that competitors cannot easily replicate. While competitors may add AI features to laboratory workflows, they cannot scale a model of care they do not have.

The requested decision authorizes Sanguine, in accordance with the submitted implementation plan, to achieve this objective. The program incorporates mitigation measures for all identified project risks, as the goal is not merely to deploy AI more rapidly, but to render Sanguine clinically usable, governable, and consistent with Mayo’s standard of care, without compromising the institution’s reputation. As this is a cutting-edge project, we remain subject to unforeseen conditions inherent to the solution. Consequently, our strategy of eliminating all identified risks entails higher costs, for this reason, we have opted not to present alternative plans that would expose known risks, however minor.

We ask executive leadership to authorize three specific actions:

- \$7.1M implementation budget for the recommended Sanguine program.
- Institutional capacity for the critical enabling work: controlled access to clinical and laboratory systems, IRB preparation, PHI governance, and physician reviewer participation in clinical validation.
- Initial scope of deployment: a private pilot with 1–10 partner providers, followed by a controlled US-only launch for 3 selected test types, subject to successful Integration & V&V.

The decision facing leadership is not whether Sanguine constitutes a modern AI concept in light of current competition. The decision lies in determining whether Mayo Clinic is prepared to fund and allocate the resources necessary for the implementation path required to make the clinical interpretation of laboratory tests safe, traceable, scalable, and consistent with the Mayo Clinic Model of Care.